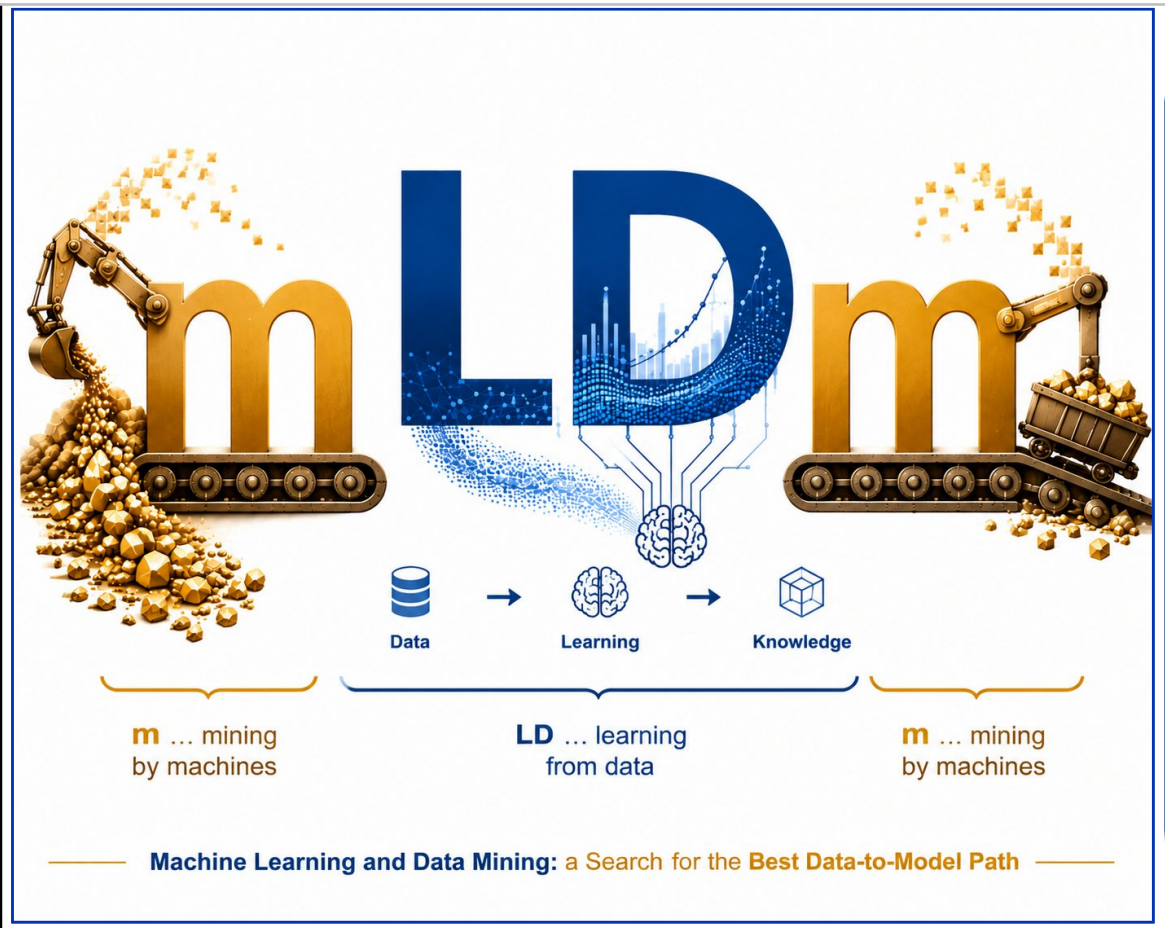
Data Source and Analytical-Dataset

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From a data source to a (defensible) analytical-dataset

in this context, “**analytical-dataset**“ means:

“**a dataset, whose observations, variables and target are defined for a specific problem**”



a dataset is not “just found”

it is constructed

from: one or more data sources

to: address a specific problem

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Three layers not to be “mixed” (or confused)

data source | *“the foundation”*

- original files, tables, events, measurements
- produced by system, collected by process

our focus, for now

analytical-dataset | ***construction***

- problem-oriented observations with defined **row meaning**, **candidate features** and,
- when needed, a **target**

modeling-dataset | *representation*

- encoded, scaled, split or otherwise transformed representation,
- supplied to a learner

*the analytical-dataset fixes the **semantic construction***
*the modeling-dataset fixes the **representation used by algorithms***

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... from the data source to an analytical-dataset

level	meaning	typical role
data source	raw tables, files, events or measurements	original or collected form
prepared data	cleaned, matched, filtered or enriched data	**intermediate working form**
analytical-dataset	one row per analytical observation for a specific problem	input to analysis or learning

- the (raw) data source may require some initial **preparation**
- such prepared data provides a **usable basis** for further dataset construction
- then, a **specific problem** provides the context for defining the **questions** to be addressed
- those questions determine **which observations and variables are selected** from prepared data

so, different questions may lead to different datasets... from the same data source(s)

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... analytical-dataset *versus* modeling-dataset

level	meaning	(possible) content
analytical-dataset	problem-oriented table or relation built from sources	all rows and attributes considered relevant
modeling-dataset	actual representation used by a learner	possibly transformed, encoded, filtered or split data
relationship	*the modeling-dataset is usually derived from the analytical-dataset*	*not every analytical variable must enter the final learner*

*the distinction matters because some decisions belong to **dataset construction**, while others belong to the **dataset representation** to be consumed by algorithms*

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A (more) formal definition of an analytical-dataset — minimum ingredients

a **task, T,** specifies,

"***the type of computational problem to be solved, e.g., classification, regression, clustering***"

an analytical-dataset, D, for a specific problem and task, T, should define:

- an observation unit or **row meaning**
- a set of attributes, X, as **explanatory** data
- an optional **target**, y, for supervised tasks
- a reproducible **construction path**, P

here D denotes the **dataset specification** not its materialized relation

using symbols, for a task T

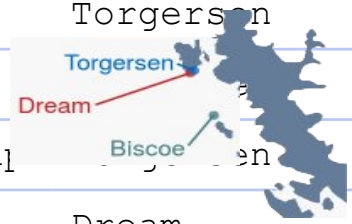
- $\mathrm{D} = (\mathrm{U}, \mathrm{X}, \mathrm{y}, \mathrm{P})$
- U = observation unit
- X = feature set
- y = target variable (if any)
- P = construction path of D

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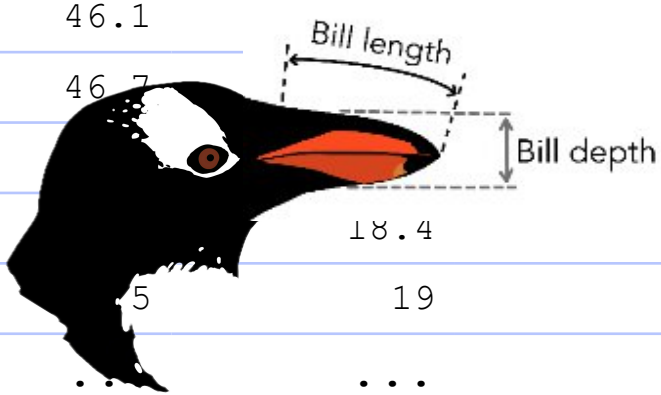
An illustrative example — the Palmer-Penguins dataset

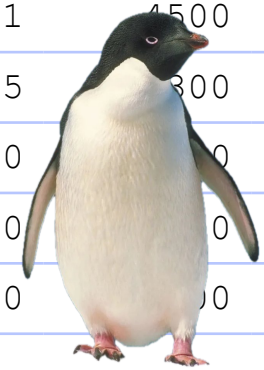
data was collected as part of research to study the Palmer Archipelago (Antarctica)
***penguins**' foraging behavior and its relationship with environmental variability*

species	island	bill length_mm	bill depth_mm	flipper length_mm	body mass_g	sex	year
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie		<null>	17.9	190	3325	female	2009
Chinstrap		52	19	197	4150	male	2007
Adelie	Dream	36.8	18.5	<null>	3500	female	<null>
Adelie	Dream	41.5	18.5	201	4000	male	2009
Gentoo		46.1		211	4500	female	2007
		46.7		195	3300	female	2007
				190	0	female	2008
			18.4	200	0	female	2008
Chinstrap	Biscoe	5	19	200	0	male	2008
...	...	...	...	...	...	...	...









adapted sample inspired by Palmers-Penguin — cf., original at www.kaggle.com

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... why this illustrative example?

strengths

- small enough to inspect manually
- mix of numeric and categorical variables
- clear target candidates

limitations

- limited metadata on collection process
- not a multi-source system
- simple grain

additional value

- good to introduce **row meaning**
- good bridge to **multidimensional** analysis
- good for **leakage** discussion

the simplicity of Palmer-Penguins is a strength
the methodological issues become visible without overwhelming technical complexity

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(a first) Formalization of the Penguins problem

problem: ***predict the species of a penguin***

element	definition
task	supervised multi-class classification
target	`species`
candidate **explanatory** variables	`island, sex, year, bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g`
candidate **observation unit**	one sampled penguin observation

this is not yet the final dataset

*it is the **first** formal **candidate path** for the modeling problem*

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(some) questions about the “technical context” of the data source

context item	question
origin	where does the data come from?
schema	which fields, types, keys and constraints exist?
extraction date/time	when was this copy or view obtained?
covered period	which temporal interval does the source describe?
filters / scope	which entities, events or records are included or excluded?
known quality issues	what completeness, consistency or validity problems are known?

using a source requires enough information to interpret and reproduce it
without its technical context, a source cannot be interpreted or reproduced

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The data source of the Palmer-Penguins

data source (schema)

- `penguins.csv`
- one row per recorded penguin observation
- has numeric and categorical attributes

candidate target

- `species`
- nominal attribute
- values `Adelie`, `Chinstrap`, `Gentoo`

why is this a “simple” source?

a compact relation for controlled analytical-dataset construction

- single relation
- simple grain
- limited provenance
- limited operational context

the example is small enough to inspect manually
while still requiring explicit row meaning, attribute roles and construction decisions

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Inspect the data source relation

```
import duckdb
```

inspect schema

```
duckdb.sql("""
DESCRIBE
FROM read_csv_auto('penguins.csv')
""").show()
```

inspect data

```
duckdb.sql("""
SELECT *
FROM read_csv_auto('penguins.csv')
LIMIT 10
""").show()
```

why to inspect?

inspect precedes dataset construction

- check field names
- check inferred types
- “scan” a few records
- search for identifiers, context attributes and measurements
- detect mismatches between the source and the analytical question

a source is used through explicit access statements
not through an implicit “dataset already exists” assumption

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The observation-unit and the data-grain

observation-unit

what **one row** in the analytical dataset is intended to **represent**

data-grain

the **level-of-detail** at which data or aggregation is represented

penguins

- **observation-unit** (the “one row” meaning) = one sampled penguin observation
- **data-grain** (level-of-detail) = same detail of penguin observation, i.e., no aggregation

example of data-grain mismatch

joining **one** penguin observation with **multiple** finer-grain event/measurement rows **duplicates the observation unit**, unless those rows are aggregated first

*a rule that applies **before combining sources or constructing features**,*
*is that **all fields must be aligned with the intended observation-unit***

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(a summary of) The data source attributes and their candidate roles

attribute	type	candidate role	remark
`species`	categorical nominal	target	class to predict
`island`	categorical nominal	feature / context	sampling location
`sex`	categorical nominal	feature / individual	biological characteristic
`year`	temporal / categorical ordinal	feature / context	collection period
`bill_length_mm`	numerical	feature / measure	physical measurement
`bill_depth_mm`	numerical	feature / measure	physical measurement
`flipper_length_mm`	numerical	feature / measure	physical measurement
`body_mass_g`	numerical	feature / measure	physical measurement

now, the same source field is interpreted through a modeling role

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Inspect — missing values

row eligibility begins with quantified source evidence

inspect missing values

```
SELECT
    COUNT(*) AS n_rows,
    COUNT(*) - COUNT(island) AS missing_island,
    COUNT(*) - COUNT(body_mass_g) AS missing_body_mass_g,
    COUNT(*) - COUNT(bill_length_mm) AS missing_bill_length_mm
        ...
FROM penguins
```

construction decisions and implications

- quantify missing values and retain or exclude rows
- ... row exclusion (deletion) may impact class balance
- ... missing values may affect candidate feature sets

missing values are not only a cleaning issue

they affect which observations and fields are admissible in a candidate path

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Inspect — target structure

class distribution is part of problem definition and evaluation planning

```
SELECT
    species,
    COUNT(*) AS n
FROM penguins
GROUP BY species
ORDER BY n DESC
```

why it matters?

- class frequencies establish a reference baseline for the classification problem difficulty
- strong class imbalance can make accuracy misleading on its own
- the target distribution may influence the choice of evaluation metrics and splitting strategy
- target inspection should therefore precede learner selection and model comparison

a predictive task is not fully specified by naming y

the distribution and meaning of y must be inspected

Inspect — multidimensional perspective

dimensions and measures

- **dimension** — an axis used to group, segment or compare observations
- **measure** — a quantitative value, typically computed via aggregation

dimensions?

- `species`
- `island`
- `sex`
- `year`

(some) **measures?**

- `COUNT(*)`
- `AVG(body_mass_g)`
- `AVG(flipper_length_mm)`
- `COUNT(*) BY year`

a **dimension** may also be used to define how a **measure** is computed

purpose?

- compare target classes across several dimensions
- assess if attributes jointly separate target classes
- form hypotheses about useful feature combinations

this is a user-driven analytical navigation — it does not replace learning
it helps to reveal structure that can guide learned construction and evaluation

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... one query, several aggregation levels — `GROUP BY CUBE`

`CUBE` supports multidimensional inspection
without writing one separate `GROUP BY` query for each aggregation level

inspect multiple dimensions

```
SELECT
    species,
    island,
    sex,
    COUNT(*) AS n
FROM penguins
GROUP BY CUBE(species, island, sex)
ORDER BY species, island, sex;
```

the behavior of CUBE

- generate the family of groupings induced by the selected dimensions
- ... it includes detailed cells, partial totals and the grand total

it enables a systematic comparison of measures on multiple dimension combinations

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The output of a `CUBE`

the meaning of `NULL` in the output of `CUBE`

species	island	sex	*meaning*
Adelie	Torgersen	female	**fully specified cell**
...	...	...	
Adelie	Torgersen	**NULL**	**`Adelie` × `Torgersen`, aggregated over `sex`**
...	...	...	
Adelie	**NULL**	**NULL**	**`Adelie` (only), aggregated over `island` and `sex`**
...	...	...	
NULL	**NULL**	**NULL**	**grand total**

`CUBE` *gives a formal multidimensional summary*

`NULL` *is usually an aggregation marker, not necessarily a missing source value*

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Identification of the aggregation levels

***displays** the 3 grouping flags one bit per dimension*

```
bitstring( GROUPING_ID(species, island, sex)::BIT, 3 )::VARCHAR AS bits
```

the "bit rule"

0 = the dimension is **used** as a grouping level

1 = the dimension is **aggregated** over all its values

bits	species	island	sex	*meaning*
000	used	used	used	species × island × sex
001	used	used	aggregated	species × island
011	used	aggregated	aggregated	species
111	aggregated	aggregated	aggregated	grand-total
010	used	aggregated	used	species × sex
...				

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... from the CUBE, select one analytical level (view)

in this context, "**analytical view**" means:

"**a selected aggregation level organized to answer a specific question**"

```
SELECT
    species,
    island,
    sex,
    COUNT(*) AS n
FROM penguins
GROUP BY CUBE(species, island, sex)
HAVING bitstring(
    GROUPING_ID( species, island, sex)::BIT, 3 )::VARCHAR = '001'
ORDER BY species, island;
```

the **bitmask** makes CUBE levels selectable

selecting a CUBE level defines an explicit, reproducible **analytical view**

species	island	sex	n
Adelie	Biscoe	NULL	...
Adelie	Dream	NULL	...
Adelie	Torgersen	NULL	...
Chinstrap	Dream	NULL	...
Gentoo	Biscoe	NULL	...
...			

meaning

- bits = '001' selects the aggregation level where only sex is rolled up
- ... result has one row per species × island

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... include analytical measures

counts can be combined with numerical summaries

inspect multiple dimensions and measures

```
SELECT
    species,
    island,
    sex,
    COUNT(*) AS n,
    AVG(body_mass_g) AS avg_mass,
    AVG(flipper_length_mm) AS avg_flipper
FROM penguins
GROUP BY CUBE(species, island, sex);
```

inspect questions

- which groups have too few observations for reliable analysis?
- do species differ across morphological measures?
- does island add variation within species?
- ...

measures *provide descriptive evidence for candidate feature hypotheses*

dimensions *define the groups over which measures are computed*

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The pivoted analytical view — a useful tool

inspect pivoted data

```
SELECT *
FROM (
    SELECT island, species, COUNT(*) AS n
    FROM penguins
    GROUP BY CUBE(island, species, sex)
    HAVING GROUPING_ID(island, species, sex) = 1
) t
PIVOT (
    SUM(n)
    FOR species IN ('Adelie', 'Chinstrap', 'Gentoo')
    GROUP BY island
);
```

turning grouped results into a comparison table

island	Adelie	Chinstrap	Gentoo
Biscoe	...	...	...
Dream	...	...	...
Torgersen	...	...	...
...			

why useful?

- compact comparison across groups
- exposes missing combinations
- useful for summaries and reports
- ...

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From inspection to candidate hypothesis

a candidate feature set encodes a testable modeling hypothesis

H1: body measurements are predictive

- ***question***: can body measurements predict species?
- ***use***: bill-length, bill-depth, flipper-length and body-mass

H2: context provides additional predictive information

- ***question***: can contextual variables improve species prediction?
- ***use***: island and year (i.e., geographic and temporal context)

H3: sex provides additional predictive information

- ***question***: can sex improve species prediction beyond body measurements?
- ***use***: sex (an intrinsic individual characteristic — different from the above)

H4: prediction improves by combining the information from H1, H2 and H3

Candidate analytical-dataset specification

a path becomes concrete when a dataset construction, $\mathrm{D} = (\mathrm{U}, \mathrm{X}, \mathrm{y}, \mathrm{P})$, for task, T, is specified

element	specification
T: task	supervised multi-class classification
U: observation unit	one sampled penguin observation
X: feature set (cf., H4, previous slide)	`island, sex, year, bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g`
y: target variable	`species`
P: construction path of D	from `penguins` select X and y; retain rows with non-missing values

this specification is a candidate analytical-dataset path
it can be revised or compared with alternatives

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Materializing the analytical-dataset

```
CREATE OR REPLACE TABLE dataset_H4 AS
SELECT
    bill_length_mm,
    bill_depth_mm,
    flipper_length_mm,
    body_mass_g,
    island, year,
    sex,
    species
FROM penguins
WHERE bill_length_mm IS NOT NULL
  AND bill_depth_mm IS NOT NULL
  AND flipper_length_mm IS NOT NULL
  AND body_mass_g IS NOT NULL
  AND island IS NOT NULL
  AND year IS NOT NULL
  AND sex IS NOT NULL
  AND species IS NOT NULL;
```

the specification defines a **reproducible** relation over a **fixed data source snapshot**

construction properties

- selected fields are clearly defined
- row inclusion follows stated criteria
- construction logic is deterministic
- dataset specification is reproducible
- path from source to dataset is auditable

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The data-leakage — a formal view

information not-available at prediction time is inadvertently used during learning

leakage-type	definition	typical example
target	features contain information derived form target	a feature that is a synonym of the target
temporal	features use information from after prediction time	tomorrow's value used to predict today's
pre-processing	transformations fitted with data outside training set	global data scaling before train/test split
train-test	test data used directly or indirectly to fit the model	data from test used in training
dependency	related observations split across train and test data	same entity in both sides

data-leakage is not only a learner problem
it may arise in data source use, dataset construction, representation or validation

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(some) data-leakage and deploy-risk scenarios — safe, unsafe or depends

scenario	safe?	reason
X includes `body_mass_g`	yes	`body_mass_g` is an explanatory feature
X includes `species`	no	target is directly present
X includes code derived from `species`	no	target is indirectly encoded
scale with full data before splitting	no	test data influences transformation parameters
X includes `year`	***depends***	risky if `year` only reflects the data collection period and prediction extend beyond that period

the label “depends” is not ambiguity
it means that the deployment semantics must justify the variable

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(other) data-leakage scenarios — pre-processing

pre-processing — **safe**

split full-dataset in train and test

- **compute** transformation parameters using **only the train** data
- **apply** parameters to **train and test**

pre-processing — **unsafe**

use full-dataset

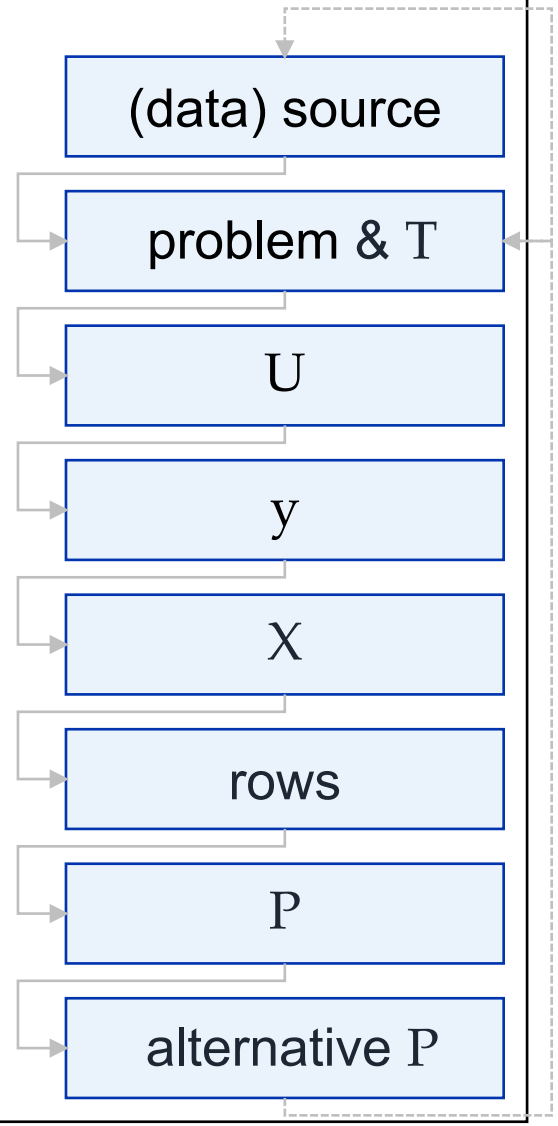
- compute transformation parameters
- transform all rows
- split into train and test

fit pre-processing on the current rtaining partition only
apply the fitted transformation unchanged to testing data

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An analytical-dataset construction checklist and a “workflow”

1. what is the (data) source and its technical context?
2. what problem is the dataset intended to address?
3. what is the observation unit and grain?
4. what is the target, if any?
5. which source attributes become candidate features?
6. is each candidate feature available at prediction time?
7. which rows are eligible for inclusion?
8. where can leakage enter the construction path?
9. is the construction specification deterministic and reproducible?
10. which alternative construction path should this be compared with?



... and, after the analytical-dataset — the modeling-dataset

construction decisions and representation decisions are related but different

construction decisions	representation decisions
analytical-dataset	**modeling-dataset**
• task-oriented relation • selected attributes and retained rows • ... *attributes*, e.g., `island, year, bill_length_mm, species, ...`	• representation supplied to an algorithm • encoding, scaling, normalization, ... • ... *objects*, e.g., `X_train, y_train, X_test, y_test`

for example, encoding and scaling may change the geometry "seen" by the learner but, they do not change the semantic meaning of the source attributes

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A final “one sentence” message



*an analytical-dataset is **constructed**, not simply found!*

*its observation-unit, features and target must be explicitly **defined***

its construction must be **traceable** to the **source** and **problem**

a **defensible dataset** must be **reproducible and comparable**